Deep Contrastive Clustering for Signal Deinterleaving

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In a complex electromagnetic environment, radar signal deinterleaving (RSD) is a challenging task. In this article, a deep contrastive clustering algorithm (DCCA) is advanced in a new self-supervised paradigm for the accurate RSD without any prior information about radar emitters. First, a contrastive self-supervised deep attention network (CSDAN) is constructed to learn signal representations by using self-defined pseudolabels of augmented signals as supervision. We use CSDAN to learn the differences between different radiation source data and generate deep features suitable for clustering. Three metrics are then used to automatically determine the number of clusters for the subsequent clustering. Extensive experiments are performed on several datasets containing different numbers of emitters. The results show that the proposed DCCA can accurately determine the number of emitters and deinterleave radar pulses. Furthermore, CSDAN can extract discriminative features of emitters with low intraclass similarity and high interclass similarity.

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I. INTRODUCTION

RADAR signal deinterleaving (RSD) aims to separate the interleaved pulses belonging to different radar transmitters, which plays an important role in modern electronic warfare [1]. RSD can be reduced to a representative clustering task [2], [3]. Over the past decades, many efforts have been made in RSD, which can be mainly categorized into traditional pulse parameter-based methods [4] and machine-learning-based methods [5].

In the first category, the inherent regularity existing in the time-of-arrival (TOA) sequence difference is exploited to separate pulses from multiple radar sources. For example, the cumulative difference histogram [6], the sequential difference histogram [7], and the repetition interval (PRI) transform [8] are proposed by analyzing the first-order difference of TOA (DTOA) sequence. The extracted pulse parameters are used for retrieval in a known radar database. Besides TOA and PRI, pulsewidth and radio frequency are jointly used as pulse description words for more effective deinterleaving [4]. Other pulse parameters are studied later, including the second-order DTOA [9], the main-lobe pulse correlation [10], Gini impurity [11], semantic codes [12], Markov process [13], and Kalman filter [14]. Although these features have clear and intuitive theories, their retrieval requires the prior knowledge of the radar transmitter, which is not always available in an uncooperative electromagnetic environment.

In recent years, machine learning has been introduced into RSD, thus formulating another category of RSD. In these methods, the pulse features are first extracted and then clustered into several groups by some clustering algorithms [15], such as K-means clustering [16], density-based spatial clustering (DBSCAN) [17], fuzzy clustering [18], ordering points method [19], agglomerative nesting [20], support vector clustering [21], self-organizing feature maps neural network [22], density dynamic clustering [23], and so on.

In the machine-learning-based RSD method, the clustering results rely heavily on the pulse features. A lot of work has been done on machine-learning-based RSD. According to the generation of pulse features, they can be mainly categorized as expert empirical features [4] and machine-learning features [5]. Empirical features are carefully designed by experts, so the performance of these methods depends heavily on the experience of the experts. Machine-learned features usually perform feature extraction [5] or feature selection [24] on pulses or empirical pulse parameters. For example, Chen et al. [25] proposed to use latitudinal bispectral integration to extract fingerprint features and combine them with clustering for individual recognition. Satija et al. [26] proposed a fingerprint feature extraction method called variational-mode decomposition and spectral features and used Taylor polynomials to model transmitter amplifiers. Kawalec and Owczarek [27] identified linear frequency modulation of radiation source signals as an important feature and proposed a feature parameter selection strategy. However, they suffer from high computational cost because some dimensionality reduction algorithms and iterative optimization algorithms are often used.

In very recent years, deep neural networks (DNNs) have been used for automatic feature learning and RSD [28], [29], [30], [31], [32], [33], [34]. In most of these works, different types of DNNs are used to learn discriminative features from some labeled pulses. For example, semantic segmentation is used to deinterleave by marking each pulse in the pulse streams according to the category of semantics it contains [28]. In [29], a pulse deinterleaving method is proposed based on implicit features learned from long short-term memory networks and statistical analysis [29]. In [30], DNN is used to separate the raw baseband mixture signals in the time domain and to deinterleave signals from the received mixture in a supervised manner [30]. In [31], a complex DNN is combined with density peak clustering for deinterleaving multiple radar signals. It uses a limited penetrable visibility graph to construct the network from interleaved radar pulse sequences and then uses a label propagation algorithm to detect community structures. In [32], the features of 2-D time PRI images are extracted by a convolutional neural network (CNN) for RSD. A recursive deinterleaving algorithm is proposed based on blind signal separation, and deep learning is proposed where the recursive deinterleaving network of deep TOA mask maps the TOA train to an appropriate feature space [33]. In [34], multiparameters are used in a single step in the employed bidirectional gated recurrent unit.

Although machine-learning-based RSD methods usually outperform traditional methods, they often require a lot of supervised information to perform well in complex electromagnetic environments. On the other hand, the number of clusters needs to be predefined. In this article, a contrastive self-supervised deep attention network (CSDAN) is constructed to learn signal representations in a self-supervised paradigm. A deep Siamese convolutional module with squeeze and excitation (SE) attention is designed to extract discriminative features of radar pulsed for RSD. Compared with available supervised DNNs for RSD [28], [29], [30], [31], [32], [33], [34], CSDAN operates in a novel self-supervised paradigm that can enhance pulse features by contrastive learning from augmented samples. In the training, the contour maps of the ambiguity function (AF) of two radar pulses are fed into CSDAN for contrastive self-supervised learning by using self-defined pseudolabels of the augmented signals as supervision. Subsequently, a deep contrastive clustering algorithm (DCCA) is developed for accurate RSD without any prior information about the radar transmitters. The learned features are shown to be effective for blind clustering. The contributions of our work can be summarized as follows.

1) A new CSDAN is constructed for RSD in a self-supervised paradigm. To the best of our knowledge, this is the first self-supervised DNN for RSD.
2) A deep Siamese convolutional module with SE attention is designed to automatically learn discriminative signal features via signal augmentation and contrastive learning. RSD can, thus, be implemented without supervision and without prior information from the radar transmitters.
3) An adaptive clustering algorithm is proposed for fully blind RSD, which uses three metrics to automatically estimate the number of clusters from the learned deep features and is more applicable in an uncooperative environment.

II. Signal Preprocessing

In modern radar systems, the matched filter is often used in the receiver chain to improve the signal-to-noise ratio. The AF of a waveform represents the output of the matched filter when the specified waveform is used as the filter input [35]. For different radar transmitters, the AF can indicate the resolution capability in both the delay and the Doppler domains, which is useful for RSD [36], [37].

The complex analytic form of the radar signal is denoted as follows:

$$\begin{aligned} s(t) &= g(t)e^{j(2\pi f_0 t+\varphi_0)} \\ &= a(t)e^{j\theta(t)}e^{j(2\pi f_0 t+\varphi_0)} \end{aligned} \tag{1}$$

where $g(t)$ is the complex envelope function of the signal waveform, f_0 is the signal carrier frequency, φ_0 is the initial phase, $a(t)$ is the amplitude modulation function, and $\theta(t)$ represents the phase modulation function. Mathematically, the AF is the square of the 2-D cross-correlation function module of the radar signal and its echo signal [38]. Therefore, the AF of the radar emitter signal is defined as follows:

$$\chi(\eta,\xi) = \int_{-\infty}^{\infty} g(t)g^*(t+\eta)e^{j2\pi\xi t}dt \tag{2}$$

where $g^*(t)$ is the conjugate function of $g(t)$, η is the time delay between the transmitted signal and the radar echo, and ξ is the Doppler frequency shift of the target moving relative to the radar. From the definition, we can also see that AF is symmetric about the origin of the coordinates. The corresponding value at the coordinate origin is the peak value of the AF, and the maximum value is $4E^2$, expressed as follows:

$$|\chi(\eta,\xi)| \le |\chi(0,0)|^2 = (2E)^2 \tag{3}$$

where E is the energy of the signal. The sections of the AF along the time axis and Doppler shift axis are equal to the autocorrelation of the complex envelope and the spectral envelope, respectively

$$\chi(\eta,0) = \int_{-\infty}^{\infty} g(t)g^*(t+\eta)dt \tag{4}$$

$$\chi(0,\xi) = \int_{-\infty}^{\infty} |g(t)|^2 e^{j2\pi\xi t}dt. \tag{5}$$

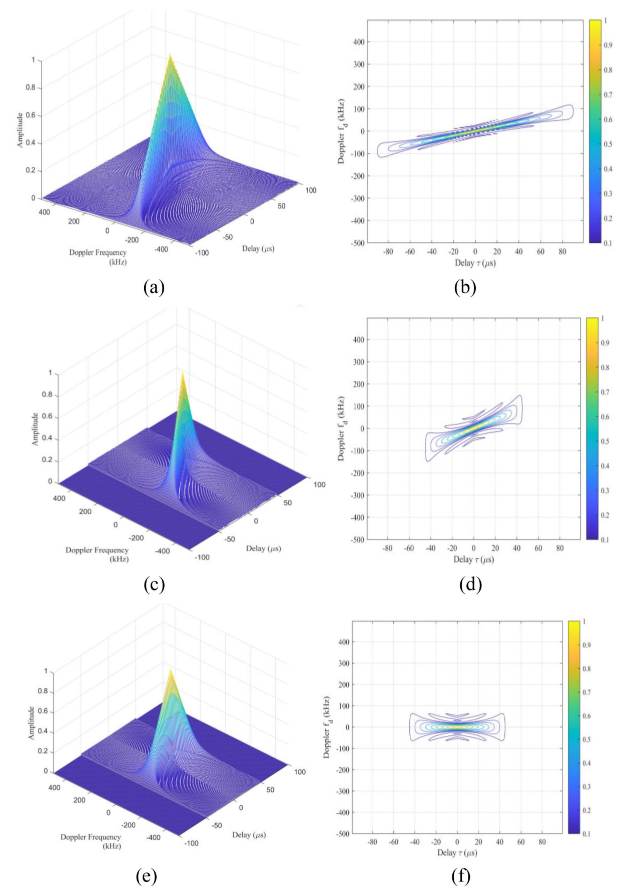

Fig. 1. AF of radar emitters. (a), (c), and (e) are the 3-D plots of AF of three radar emitters, respectively. (b), (d), and (f) are the 2-D plots of AF of three radar emitters, respectively. (a) Three-dimensional plot of AF of radar emitter 1. (b) Two-dimensional plot of AF of radar emitter 1. (c) Three-dimensional plot of AF of radar emitter 2. (d) Two-dimensional plot of AF of radar emitter 2. (e) Three-dimensional plot of AF of radar emitter 3. (f) Two-dimensional plot of AF of radar emitter 3.

As a measure of the similarity between the radar signal itself and the echo after the time delay and Doppler frequency shift, the AF can better express the unique properties of signals for signal classification [39]. The AFs of different radar signals are shown in Fig. 1, where Fig. 1(a), (c), and (e) shows the 3-D plots of the AFs of three emitters, respectively, and Fig. 1(b), (d), and (f) shows the 2-D plots of the AFs of three radar emitters, respectively. It can be seen that there is a remarkable difference in the AFs for different emitters. However, a large amount of data has to be processed for 3-D AF. Geographical contour maps show lines of constant height, which is an important tool for describing the AF in geography. Therefore, the contour map of AF at a given height reflects the distribution and hierarchical structure of energy in the time–frequency plane and is not easily affected by noise. In our work, we use the contour map of AF to represent the radar signals.

III. CONTRASTIVE SELF-SUPERVISED DEEP ATTENTION NETWORK

It is well known that supervised learning and unsupervised learning are two basic paradigms in machine learning.

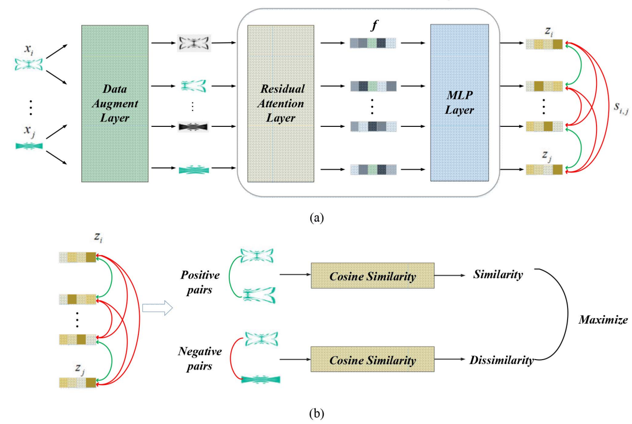

Fig. 2. Brief description of CSDAN. (a) Architecture of CSDAN. (b) Positive and negative pairs.

Unlike supervised learning, self-supervised learning artificially constructs some auxiliary tasks and corresponding supervision information, hoping to train the network and learn meaningful representation for downstream tasks through this constructed supervision information. As mentioned in Section I, in this article, we use a DNN to automatically extract signal features in a self-supervised paradigm.

A. Network Architecture

In this section, a CSDAN is constructed, as shown in Fig. 2(a). In a batch with two radar pulse samples, the contour maps of two samples are fed into the network, respectively. As shown in Fig. 2(b), the two inputs can take the form of positive and negative pairs, where positive pairs are the two augmented contour maps of pulses from the same emitter, and negative pairs are the contour maps of two pulses from different emitters. Then, some transformations, such as clipping, flipping, color dithering, and graying, are performed to obtain a pair of enhanced contour maps for the two inputs. This processing unit is called the *Data Augment Layer* in Fig. 2(a), thus formulating an auxiliary task.

In traditional CNN, the convolution operation is performed on the local area of the images and, thus, there is redundancy in the multichannel features. In CSDAN, an SE attention module is incorporated into a deep residual network ResNet18 to formulate a *Residual Attention Layer* for feature extraction of the enhanced contour maps. A detailed structure of ResNet18 is shown in Fig. 3(a), which has 18 deep layers. In the network, shortcut connections are inserted into a plain convolutional network. The identity shortcuts are used when the input and output have the same dimensions to allow for convolutional layers to work more efficiently. The structure of the ResNet block with the SE attention module is shown in Fig. 3(b). It is hoped that the model can automatically learn the importance of different channels via SE attention. Fig. 3(c) shows a detailed structure of SE. SE mainly consists of three parts: Squeeze, Exception, and Scale. The squeeze compresses the feature map via global average pooling. Then, the fully connected layer is used to predict the importance of each channel

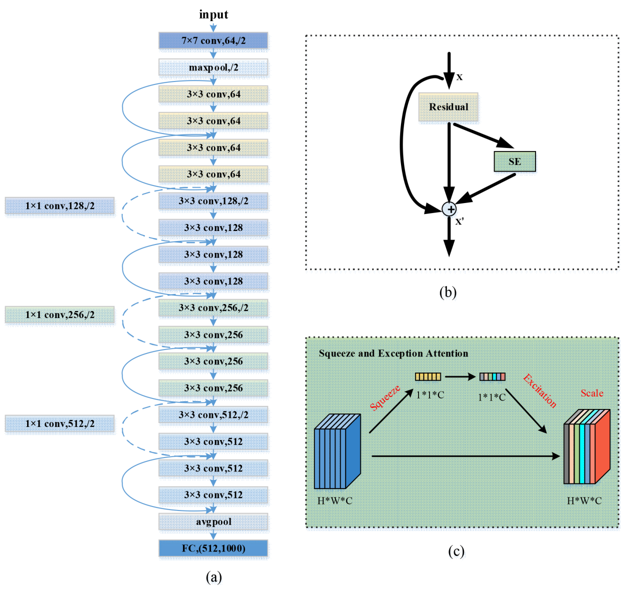

Fig. 3. Structure of the ResNet block with an attention mechanism. (a) Block structure of ResNet18. (b) Block structure of ResNet with SE attention. (c) Squeeze and Exception attention module.

through exception. Scale is a simple weighting operation on all channels to obtain the final result. Adding the attention to ResNet18 will increase the model parameters to

$$\frac{2}{r}\sum_{i=1}^{\text{num}} N_i \cdot C_i^2 \tag{6}$$

where r represents the dimensionality reduction coefficient in SE, num represents the number of stages in the residual model, C_i^2 represents the number of channels in the ith stage, and N_i represents the number of repeated blocks in the ith stage. The parameter set of Resnet18 is 11.7 M, and the SE attention module increases the parameter set by about 10%. However, the computational complexity has increased by less than 1%. Finally, a *Multilayer Perceptron Layer* follows to obtain the final representation of two inputs.

B. Contrastive Learning

Contrastive learning aims to increase the similarity between samples in the positive pairs and decrease the similarity between samples in the negative pairs [40]. In our work, contrastive learning is then performed on the CSDAN by formulating some positive and negative pairs. During training, the similarity between the two instances in the positive or negative pair is calculated from the network. The constructed network is trained to maximize the deep representations of the positive pairs while minimizing the nonlinear representations of the negative pairs. Denote the two augmented contour maps with subscripts i and j as x_i and x_j, respectively; in our work, the cosine function is used to compute the similarity $s_{i,j}$ between x_i and x_j

$$s_{i,j} = \frac{z_i^T z_j}{\|z_i\| \cdot \|z_j\|} \tag{7}$$

where z_i and z_j represent the features of x_i and x_j extracted by CSDAN, T represents the transposition operation, $s_{i,j}$ represents the similarity between x_i and x_j, and $\|\cdot\|$ represents the vector norm.

By calculating the similarity between the outputs of two branches of CSDAN and performing backpropagation training on pairs of instances, the network can learn the features that can bring similar samples closer together and different samples further apart. It should be noted that the training can be done on a separate dataset and can be easily migrated to downstream tasks. In our work, we use the unlabeled pulses from unknown emitters as the training instances. If there are some labeled samples in the downstream task, the network can be retrained by some fine-tuning strategies.

C. Loss Function

In this section, we describe the loss function of CSDAN in learning features. Compared with natural images, the difference in the contour map of the AF of different emitter signals is more subtle. In our proposed CSDAN, the normalized temperature-scaled cross-entropy loss (NT-Xent), which is derived from the *InfoNCE* loss [41], is used for training. First, all the samples are divided into several "batches" for efficient training. For any two augmented contour maps x_i and x_j in a batch, a similarity probability is calculated via a softmax function

$$\text{pro}_{i,j} = \frac{\exp(s_{i,j}/\tau)}{\sum_{k=1,k\neq i}^{2n_b}\exp(s_{i,k}/\tau)} \tag{8}$$

where i and j are the subscripts of the samples in the batch, $s_{i,j}$ represents the similarity between the contour maps x_i and x_j; τ is the temperature parameter, which is used to control the sensitivity of the loss to negative sample pairs; and n_b represents the number of samples in a batch. In NT-Xent, the loss about x_i and x_j is then defined by calculating a negative logarithm of the similarity probability

$$l_{i,j} = -\log\left(\text{pro}_{i,j}\right) \tag{9}$$

which constitutes the form of NT-Xent.

It can be seen from (9) that the loss of each pair of contour maps will be calculated repeatedly due to the position exchange. For the kth contour map in a batch, its augmented samples x_{2k-1} and x_{2k} are used to formulate a pair of samples. Therefore, the self-supervision loss of CSDAN is expressed as follows:

$$\text{Loss\_NTXent} = E\,(l) = \frac{1}{2n_b}\sum_{k=1}^{n_b}\left(l_{2k-1,2k} + l_{2k,2k-1}\right) \tag{10}$$

where E is the expectation, and $l_{2k-1,2k}$ is the negative logarithm of the probability in which the second enhanced contour map x_{2k} is most similar to the first contour map x_{2k-1} in the kth pair. After swapping the positions $2k-1$ and $2k$, we obtain $l_{2k,2k-1}$. Each pair within the batch is taken out to formulate the loss function in (10). In the training, we want the result of $l_{2k-1,2k}$ and $l_{2k,2k-1}$ to be as small as possible. After training, the learned pulse features are used for the subsequent clustering algorithms. In the experiments, $\tau = 0.25$.

IV. DEEP CONTRASTIVE CLUSTERING ALGORITHM

A. Adaptive Clustering

As mentioned in Section I, the available machine-learning-based methods require the number of clusters to be predefined, such as the K-means clustering algorithm. Some recent works have made efforts toward more efficient clustering, such as clustering-guided sparse structural learning, which exploits nonnegative spectral clustering and sparse structural learning [46] and nonnegative spectral analysis with constrained redundancy [47]. However, in these algorithms and their variants, the number of clusters in the training process is usually unknown.

It is well known that the most important thing is to determine the number of clusters and the initial cluster centers. In our method, we propose a new adaptive clustering algorithm that can automatically determine the number of clusters. In the feature space, some samples will overlap and we first approximate the cluster range $[K_{\min}, K_{\max}]$ empirically. By increasing the number of clusters K within this range, some evaluation indicators of the clustering results are calculated: adjusted rand index (ARI) [42], normalized mutual information (NMI) [43], and accuracy (AC) [44]. It is assumed that the highest indicator value k corresponds to the correct cluster number K, which is expressed as follows:

$$\max_{k \in [K_{\min}, K_{\max}]} \text{ARI}(k), \text{NMI}(k), \text{AC}(k) \tag{11}$$

where ARI(k), NMI(k), and AC(k) are the ARI, NMI, and AC indicators, respectively, of the clustering result when the number of clusters is k. This is a typical multiobjective optimization problem. Note that it is not guaranteed that the peaks of different indicators correspond to the same number of clusters. The best reallocation can be achieved using the Hungarian algorithm. In our work, a simple strategy is proposed for adaptive clustering. That is, if two indicators correspond to the same cluster number, the cluster number is considered to be the correct number of clusters.

Once the number of clusters is determined, the K-means algorithm is used for the final RSD. However, the standard K-means algorithm uses random initialization of the initial cluster centers, which can lead to different clustering results in each run. K-means++ believes that the initialized clustering centers should be as far apart as possible, which can lead to more stable clustering results. This is why, we use K-means++ in our work. This gives rise to the DCCA, the architecture of which is shown in Fig. 4. The time and space complexity of the clustering algorithm includes the determination of the cluster number and the K-means++ clustering. When determining the number of clusters, we can use a large upper bound and a small lower bound. To speed up the optimization, the number of clusters K can be roughly estimated by the learned low-dimensional feature distribution.

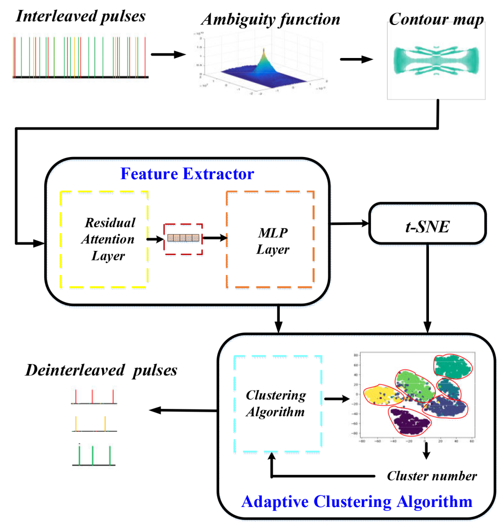

Fig. 4. Architecture of DCCA.

Since determining the number of clusters is very fast compared with clustering, it can be ignored. The time complexity and space complexity of DCCA can be expressed as $O(KI_{\max}mn)$ and $O((n + K)m)$, where K is the number of clusters, $I_{\max}$ is the maximum number of iterations, m represents the dimensionality of the learned deep features, and n is the number of pulses. Since K, $I_{\max}$, and m can be considered constants, both the time and space complexities of DCCA are $O(n)$.

B. Procedure of DCCA

1) *AF Calculation:* First, the AF of the radar pulse is calculated. The calculation of the AF is related to the time and the Doppler frequency shift of the signal. Therefore, the time range and the Doppler frequency-shift range should be estimated first. Then, the spectrum of the radar pulses is calculated, and the frequency position corresponding to the peak of the spectrum is the rough estimate of the real frequency [45]. The carrier frequency is estimated from the signal spectrum, which is given by

$$\hat{f}_c = k_m \cdot \frac{f_s}{N} \tag{12}$$

where $\hat{f}_c$ is the estimated signal frequency, k_m is the frequency position of the spectrum peak, f_s is the sampling frequency of the signals, and N is the signal length. The range of the Doppler frequency shift is determined by the difference between the estimated signal carrier frequency and its ground truth. The corresponding frequency offset Δf of the carrier frequency f_c is expressed as follows:

$$\Delta f = \hat{f}_c - f_c. \tag{13}$$

After determining the time- and frequency-shift range of the signal, the 3-D AF of each signal is calculated according to (2).

2) *Contour Map Generation:* The contour map is then generated by setting a given height of the AF. Fig. 5 shows the contour maps of AF with different heights, where

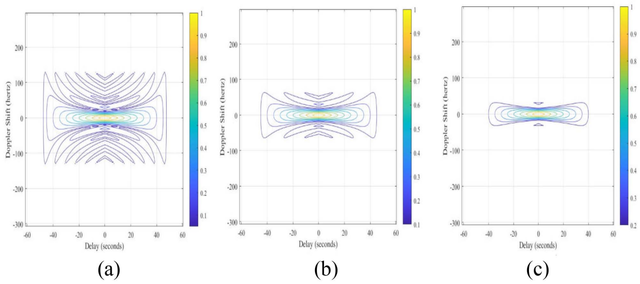

Fig. 5. Contour lines of the AF. (a) Height is set to the peak of the AF. (b) Height is set to 0.1 times of the peak of the AF. (c) Height is set to 0.3 times the peak of the AF.

Fig. 5(a)–(c) shows the results of setting the height as the peak of AF, 0.1 times of the peak of AF, and 0.3 times of the peak of AF, respectively.

In order to increase the difference in contour densities between the main AF peak and the secondary peaks, a suitable height should be determined. In our work, the height H is empirically set at 0.1 times the peak of AF, which is expressed as follows:

$$H = 0.1 * h_{\max} \tag{14}$$

where $h_{\max}$ is the value at the zero point of the AF.

3) *Feature Extraction Via CSDAN:* The contour maps of unknown radar pulses are fed into a trained CSDAN for feature extraction of pulses. The CSDAN is trained by contrastive learning, as described in Section III-B.

4) *Adaptive Clustering:* The extracted features of the radar pulses are clustered by the adaptive clustering, as described in Section IV-A. The clustering number is determined according to the indicators. Then, K-means++ clustering is performed to cluster the pulses into several groups.

V. EXPERIMENTAL RESULTS AND DISCUSSION

A. Datasets and Experimental Conditions

In this section, the public navigation radar emitter dataset in the "Radar emitter individual recognition technology competition" in 2020 is used to validate the performance of our proposed method. It contains two groups of interleaved pulses. The first group is a navigation radar signal set containing ten emitters, called Dataset I. Each radar contains 1000 pulses, and the carrier frequency is about 100 MHz, and the sampling frequency is 400 MSps. Dataset II is an interleaved navigation radar pulse set containing 40 emitters. Each radar radiation source contains 1000 pulses, and the carrier frequency is approximately 50 MHz and the sampling frequency is 200 MSps.

Three datasets with different numbers of radar emitters are constructed based on them. First, six types of emitters in Dataset I were used to form subdataset (Dataset 1) with a small number of categories. Second, 12 types of radar emitters from Dataset II were used to form a subdataset (Dataset 2) with a medium number of categories. Finally, 18 types of radiation sources in Dataset II were used to form a subdataset (Dataset 3) with a relatively large number of categories. The deep network model in this article is built using the Python framework and the software system is Linux. Several experiments are designed to verify the effectiveness and competitiveness of the proposed method. Here, the number of clusters corresponding to the highest external indicators and the clustering results are used as the cluster number for clustering. Three evaluation indicators: ARI, NMI, and AC are used to evaluate the clustering results. Training and testing of the model are performed on the 64 GB RAM HP Z840 server equipped with E5-2697v3 CPU and NVIDIA GeForce GTX 2080 Ti GPU graphics card.

B. Experiment 1: Analysis of Extracted Pulse Features

To evaluate the capability of CSDAN in feature extraction, in this section, we first analyze the learned features of different radar emitters from CSDAN and compare them with those of some related methods. In the experiment, we set the batch size $N = 12$, the number of epochs $= 200$, and the learning rate $\mathrm{lr} = 0.0003$. To stabilize the training, we use the Adam optimizer in the training. The methods we compared include the following:

1) classical feature extractors, such as principal component analysis (PCA);
2) unsupervised DNNs, such as AutoEncoder (AE);
3) variations of CSDAN.

In the second category, the AE consists of five convolutional layers in the encoding stage and five deconvolutional layers in the decoding stage. The kernel sizes of the first four layers and the last layer are 1×3 and 2×3, respectively. All of their convolutional strides are 1×2. The kernels and strides in the decoding stage are the same as those in the encoding stage except for the last deconvolutional layer, and each layer is followed by a leaky Relu activation function and a batch-normalization operation [46], [47]. In the third category, the influence of the SE attention module and the contour map of the AF on the features are discussed by constructing some variations of CSDAN. First, the SE attention module is removed from the *Residual Attention Layer* in CSDAN and compared with DCCA. This method uses the contour maps of AF as inputs and is denoted as AF-WO, where WO indicates no attention. Other parameter settings in the experiment are the same as that of DCCA in Section IV-B. In addition, different inputs are compared, including the short-time Fourier transform (STFT) of signals and the bispectrum (BS) of signals. These methods are referred to as STFT and BS, respectively. Dataset 1, Dataset 2, and Dataset 3 are used for comparison. T-distributed stochastic neighbor embedding [48] is used for dimensionality reduction to visualize the features on Dataset 1, Dataset 2, and Dataset 3, respectively, and the results are shown in Fig. 6.

Fig. 6(a)–(c) shows the learned low-dimensional feature distribution maps of the three datasets by CSDAN.

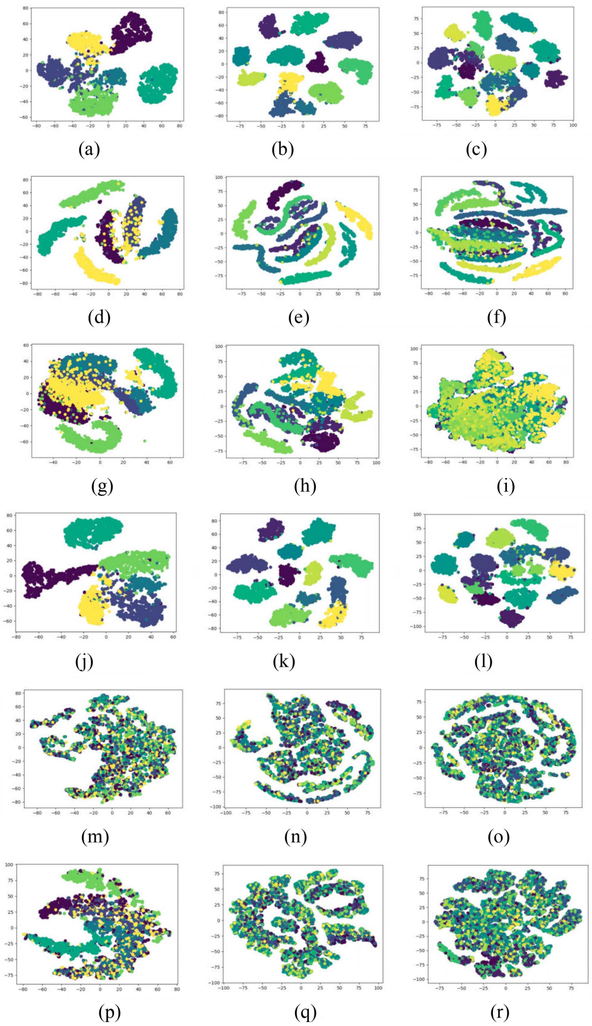

Fig. 6. Distribution of low-dimensional features on the three datasets. (a) CSDAN-Dataset 1. (b) CSDAN-Dataset 2. (c) CSDAN-Dataset 3. (d) PCA-Dataset 1. (e) PCA-Dataset 2. (f) PCA-Dataset 3. (g) AE-Dataset 1. (h) AE-Dataset 2. (i) AE-Dataset 3. (j) AF-WO-Dataset 1. (k) AF-WO-Dataset 2. (l) AF-WO-Dataset 3. (m) STFT-Dataset 1. (n) STFT-Dataset 2. (o) STFT-Dataset 3. (p) BS-Dataset 1. (q) BS-Dataset 2. (r) BS-Dataset 3.

Fig. 6(d)–(f) shows the low-dimensional feature distribution maps of the three datasets after PCA. Fig. 6(g)–(i) shows the low-dimensional feature distribution maps of the three datasets after feature extraction by AE. Fig. 6(j)–(l) shows the learned features of the three datasets by AF-WO. Fig. 6(m)–(o) shows the learned features of the three datasets by DCCA with STFT. Fig. 6(p)–(r) shows the features learned from the three datasets by DCCA with BS. Different colors represent different categories in the figures. It can be seen from the figures that CSDAN can learn discriminative pulse features on the three datasets, and the learned features have low intraclass similarity and high interclass similarity. However, the pulse features extracted by PCA, AE, and AF-WO tend to overlap between different categories, making it difficult to cluster the pulses into the correct emitter group. In addition, the learned pulse features extracted by DCCA with STFT input and BS input can hardly be distinguished.

TABLE I
Clustering Results of Dataset 1

Cluster Number K \ Indicators	NMI	ARI	AC
5	0.8197	0.7973	0.8744
6	**0.8350**	**0.8501**	**0.9240**
7	0.8025	0.7687	0.8277
8	0.7729	0.6962	0.7460
9	0.7608	0.6521	0.6860

The bold values indicate the best values in the comparative experiment.

For CSDAN, the contrastive learning can help to capture the intrinsic features of pulses for clustering. Compared with other unsupervised methods, such as PCA and AE, CSDAN is more suitable for clustering tasks where the samples are very close to each other. For the training of AE, its extract features are distributed in irregular shapes that will degrade the high degree of confusion, which has very low discriminative ability, as shown in Fig. 6(g)–(i).

Although samples from different categories are clustered into clusters by our proposed CSDAN, for the features extracted by CSDAN without attention, the features of samples from different categories tend to overlap, as shown in Fig. 6(j)–(l). The results of STFT and BS are shown in Fig. 6(m)–(o) and (p)–(r), respectively. It can be seen that both the STFT and BS of signals could not well distinguish different emitters when compared with the contour map of the AF.

C. Experiment 2: Deinterleaving Results of Different Methods

The above results show that our method can extract discriminative features of radar emitters. The learned features are then used for RSD. In this section, we compare the deinterleaving results of different clustering methods. The upper and lower bounds of the number of clusters can be determined either by setting large values or by an approximate estimation from feature maps. The evaluation indicators NMI, ARI, and AC are used to evaluate the clustering results. Then, the proposed adaptive clustering algorithm is used to determine the number of clusters K within the upper and lower bounds by increasing the number of clusters K in turn and calculating the maximum value of the cluster indicators to indicate the number of clusters. The experimental results of DCCA on the three datasets are shown in Tables I–III, respectively. The tables present the number of clusters and the corresponding indicators, and the bold indicates the true number of emitters.

The performance of our proposed DCCA can be intuitively observed from the evaluation indicators. From the tables above, we can see that DCCA can indicate the correct number of clusters of all three datasets. Furthermore, we can observe from the indicators that, as the number of clusters increases, the clustering indicators under different cluster numbers become closer. In other words, as the actual

TABLE II
Clustering Results of Dataset 2

Indicators / Cluster Number K	NMI	ARI	AC
10	0.9357	0.8941	0.8813
11	0.9540	0.9363	0.9420
12	**0.9646**	**0.9696**	**0.9826**
13	0.9504	0.9331	0.9347
14	0.9353	0.8871	0.8827
15	0.9213	0.8471	0.8349

The bold values indicate the best values in the comparative experiment.

TABLE III
Clustering Results of Dataset 3

Indicators / Cluster Number K	NMI	ARI	AC
15	0.8770	0.8110	0.8349
16	0.8904	0.8419	0.8713
17	0.8991	0.8649	0.8993
18	**0.9057**	**0.8744**	**0.9040**
19	0.9024	0.8641	0.8931
20	0.8971	0.8446	0.8704

The bold values indicate the best values in the comparative experiment.

TABLE IV
Clustering Results of Dataset 1 Without Attention

Indicators / Cluster Number K	NMI	ARI	AC
4	0.7567	0.6808	0.7586
5	**0.7849**	**0.7694**	**0.8608**
6	0.7262	0.7026	0.7764
7	0.7452	0.7182	0.8091
8	0.7178	0.6271	0.7128

The bold values indicate the best values in the comparative experiment.

number of clusters increases, so does the difficulty of the RSD task.

The clustering experimental results without the attention mechanism of the three datasets are shown in Tables IV–VI, respectively. It can be seen from Table V that when the actual number of clusters is 12, although it can be clustered to the correct number of clusters, the AC value decreases from 0.9826 to 0.9589 compared with the clustering result with the attention mechanism, and the other two indicators also decrease to some extent. When the actual number of clusters is 6 or 18, it cannot be clustered to the correct number without adding an attention mechanism to the feature extraction module. Therefore, the attention mechanism plays a very important role in the feature extraction process of DCCA by evaluating the importance of features from different channels in the network training. As mentioned above, the contour maps of the AF from different emitters are more difficult to discriminate. The attention mechanism

TABLE V
Clustering Results of Dataset 2 Without Attention

Indicators / Cluster Number K	NMI	ARI	AC
10	0.8951	0.8256	0.8586
11	0.9268	0.8975	0.9241
12	**0.9424**	**0.9293**	**0.9589**
13	0.9282	0.8907	0.9095
14	0.9139	0.8451	0.8560
15	0.9072	0.8177	0.8176

The bold values indicate the best values in the comparative experiment.

TABLE VI
Clustering Results of Dataset 3 Without Attention

Indicators / Cluster Number K	NMI	ARI	AC
15	**0.8956**	0.8452	0.8605
16	0.8866	0.8424	0.8595
17	0.8937	**0.8596**	**0.8902**
18	0.8909	0.8400	0.8579
19	0.8824	0.8072	0.8240
20	0.8766	0.7868	0.8044

The bold values indicate the best values in the comparative experiment.

TABLE VII
Clustering Results of PCA+K-Means Algorithm on Dataset 1

Indicators / Cluster Number K	NMI	ARI	AC
3	0.369	0.246	0.458
4	0.381	0.272	0.494
5	0.391	0.282	**0.500**
6	0.418	0.261	0.452
7	0.457	0.270	0.407
8	0.453	0.270	0.418
9	**0.509**	**0.323**	0.421

The bold values indicate the best values in the comparative experiment.

will help the network to focus on the slight difference, thus producing a gain in feature extraction. In addition, the attention module can reduce the dimensionality of the features and enhance important features. Thus, SE attention will also reduce the number of network parameters and computational complexity.

Tables VII to IX present the clustering results of another comparative method, PCA followed by K-means clustering. Tables VII–IX present the results of the three datasets, respectively. It can be seen from Tables VII and VIII that, in the experimental results of Dataset 1 containing 6 types of radiation sources and Dataset 2 containing 12 types of radiation sources, the low-dimensional features obtained by PCA could not be clustered to the correct cluster number, and the cluster indicators are distorted. It can be seen from Table IX that, in the experiment of Dataset 3 containing

TABLE VIII
Clustering Results of PCA+K-Means Algorithm on Dataset 2

Cluster Number K \ Indicators	NMI	ARI	AC
9	0.414	0.194	0.296
10	0.422	0.202	0.323
11	0.446	0.217	0.340
12	0.467	0.224	0.323
13	0.487	0.246	**0.364**
14	0.487	0.239	0.354
15	**0.507**	**0.251**	0.339

The bold values indicate the best values in the comparative experiment.

TABLE IX
Clustering Results of PCA+K-Means Algorithm on Dataset 3

Cluster Number K \ Indicators	NMI	ARI	AC
15	0.437	0.182	0.292
16	0.438	0.177	0.284
17	0.447	0.191	0.294
18	**0.471**	**0.210**	**0.311**
19	0.454	0.181	0.268
20	0.469	0.199	0.303
21	0.470	0.206	0.290

The bold values indicate the best values in the comparative experiment.

TABLE X
Clustering Results of Dataset 1 by AE

Cluster Number K \ Indicators	NMI	ARI	AC
3	0.509	0.393	0.483
4	0.434	0.317	0.438
5	0.566	0.463	0.605
6	**0.603**	**0.508**	**0.682**
7	0.552	0.428	0.593
8	0.543	0.411	0.543
9	0.560	0.405	0.509

The bold values indicate the best values in the comparative experiment.

18 types of radiation sources, this comparison algorithm can cluster to the correct number of clusters. However, compared with the algorithm proposed in this article, the AC value decreases by nearly 0.59, and the other two indicators also decrease significantly. It is reflected from the side that the traditional dimensionality reduction and clustering schemes could not capture the intrinsic characteristics of the emitters and cluster them.

We then compare our method with an unsupervised DNN and AE. The clustering results on the three datasets are shown in Tables X–XII, respectively. AE is trained to reconstruct the data accurately. The output of the encoder part is taken as the pulse feature for the subsequent clustering.

TABLE XI
Clustering Results of Dataset 2 by AE

Cluster Number K \ Indicators	NMI	ARI	AC
9	0.668	0.488	0.570
10	0.677	0.510	0.609
11	**0.703**	**0.535**	**0.664**
12	0.669	0.483	0.632
13	0.683	0.493	0.617
14	0.560	0.405	0.509

The bold values indicate the best values in the comparative experiment.

TABLE XII
Clustering Results of Dataset 3 by AE

Cluster Number K \ Indicators	NMI	ARI	AC
15	0.271	0.119	0.240
16	0.272	**0.121**	**0.246**
17	0.268	0.112	0.236
18	0.268	0.112	0.239
19	0.269	0.113	0.235
20	0.270	0.109	0.227
21	**0.274**	0.112	0.232

The bold values indicate the best values in the comparative experiment.

From the results, it can be observed that when the features learned by AE are used to estimate the number of clusters and cluster emitters, the true number of clusters cannot be correctly estimated. It can be seen from Table X that, in the experiment on Dataset 1 containing six types of radiation sources, different indicators can accurately indicate the true number of clusters, but there is a gap of more than 0.23 between each indicator and the indicators of our method. It can be seen from Table XI that, in the experiment on Dataset 2 containing 12 types of radiation sources, although all the indicators reach the peak at the same cluster number, the cluster number is incorrectly estimated. Table XII presents that, in the experiment on Dataset 3 containing 18 types of radiation sources, not only do the evaluation indicators change with the number of clusters but also the number of clusters indicated by different indicators is not the same. This shows that the discriminability of the features extracted by AE is poor.

D. Experiment 3: Adaptive Clustering

It should be noted that lower bounds on the number of clusters can be obtained either by manual estimation or by automatic search over a wide range. The estimated number of clusters K can then be increased in turn. In this test, we take a larger range and calculate the objective metrics of the clustering results as the number of clusters changes. The NMI, ARI, and AC of the clustering results are calculated, and the curves are shown in Fig. 7. The results of CSDAN on Dataset 1, Dataset 2, and Dataset 3

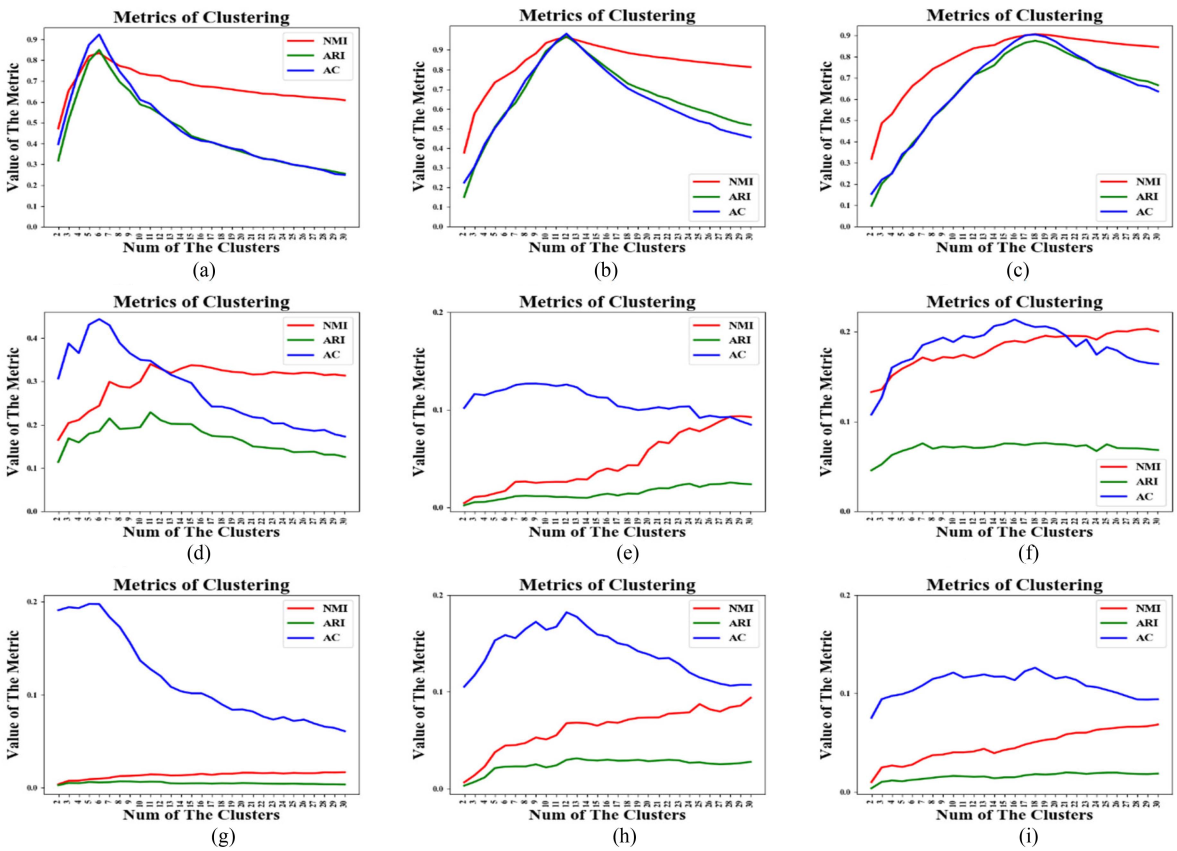

Fig. 7. Clustering metrics of different methods. (a) DCCA-Dataset 1. (b) DCCA-Dataset 2. (c) DCCA-Dataset 3. (d) STFT-Dataset 1. (e) STFT-Dataset 2. (f) STFT-Dataset 3. (g) BS-Dataset 1. (h) BS-Dataset 2. (i) BS-Dataset 3.

are shown in Fig. 7(a)–(c), respectively. From the results, we can observe that when the number of clusters is closer to the true number of clusters, the metric values are higher, and the metrics reach the peak when they reach the true number. This means that DCCA can correctly predict *K*, which appears in the correct number of clusters, even over a large range. At the same time, the NMI, ARI, and AC curves all reach the peak at the position of the true cluster number, which conforms the effectiveness of the proposed DCCA. It can be seen from Fig. 7(d)–(i) that STFT and BS cannot produce good results on the three datasets. Their clustering metrics are all distributed around 0.2, and they cannot be clustered to the true number of clusters because the STFT and BS of different transmitters are very similar, especially when the transmitted signals have the same modulation type and parameters. The datasets in this article contain only one modulation type and similar parameters, so the STFT and BS of the transmitters have very subtle differences.

E. Experiment 4: Comparative Experiments and Analysis

In this experiment, we compare our proposed method with some related works, including methods that combine traditional feature extraction with a clustering algorithm and DNN with a clustering algorithm. PCA, AE, DBSCAN, deformable convolutional neural network (DCN) [51], and deep embedded clustering (DEC) [52], [53] are taken as the comparative methods. PCA and AE use a K-means clustering algorithm for RSD by predefining a correct number of clusters. DCN and DEC jointly learn emitter features and cluster signals. The NMI, ARI, and AC of the RSD results are calculated by taking the average of 30 independent experiments. The comparison results on Dataset 2 are shown in Table XIII. In addition, the metrics of the above six methods are compared on the other two datasets.

TABLE XIII
Comparative Results on Dataset 2

Methods \ Indicators	NMI	ARI	AC
PCA	0.467	0.224	0.323
AE[48]	0.669	0.483	0.632
DBSCAN[17]	0.915	0.872	0.890
DCN[51]	0.575	0.335	0.462
DEC[52-53]	0.634	0.384	0.464
DCCA	**0.9646**	**0.9696**	**0.9826**

The bold values indicate the best values in the comparative experiment.

The comparison diagram is shown in Fig. 8. From left to right, the experimental results on Dataset 1, Dataset 2, and Dataset 3 are shown in Fig. 8(a)–(c), respectively. From the results, we can observe that our proposed DCCA remains

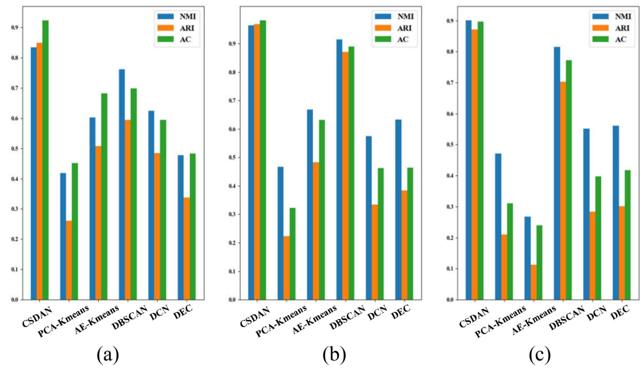

Fig. 8. Clustering indicators of the three datasets.

competitive with its counterparts. By using a deep Siamese convolutional module with SE attention to learn signal representations, DCCA can extract more discriminative features of radar pulses in a self-supervised paradigm.

VI. CONCLUSION

This article proposes a new DCCA for RSD under the premise of self-supervised to overcome the limitations of current deinterleaving methods and provide assistance for electronic reconnaissance. Several experiments are carried out on several datasets containing different numbers of radar emitters. The results show that the proposed DCCA can extract discriminative features of emitters with low intraclass similarity and high interclass similarity for fully blind RSD. In future work, we will continue to explore a more efficient network structure for RSD with a very large number of emitters.

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